Forest Fire Smoke Detection System Using Convolutional Neural Networks and Transfer Learning

1. **Project Title**

Forest Fire Smoke Detection System Using Deep Learning-Based Image Classification

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2. Background and Motivation

Forest fires are among the most destructive natural disasters, causing severe environmental damage, biodiversity loss, economic losses, and threats to human life. Early detection of forest fires is critical for reducing damage and enabling rapid emergency response.

Traditional fire monitoring systems often rely on human observation, satellite imagery, or sensor networks, which may be expensive, slow, or limited in coverage. Recent advancements in Artificial Intelligence (AI) and Computer Vision have enabled automated detection of smoke and fire from images captured by cameras, drones, and surveillance systems.

Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in image classification tasks and can effectively identify smoke patterns in forest environments. This project aims to develop a CNN-based Forest Fire Smoke Detection System and compare its performance with transfer learning models to determine the most effective approach for early wildfire detection.

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3. Problem Statement

The objective of this project is to develop an image classification system that automatically detects smoke in forest images.

The system should accurately classify images into the following categories:

- Smoke Present
- No Smoke Present

The developed model should achieve high classification accuracy while minimising false alarms and missed detections. The system should also be robust under different environmental conditions, such as varying lighting, weather, and smoke density levels.

4. Dataset Information

Dataset Name

Forest Fire Smoke Dataset

Dataset Sources

- Kaggle Forest Fire Dataset: https://www.kaggle.com/datasets
- Roboflow Forest Fire Smoke Dataset: https://roboflow.com

Dataset Description

The dataset contains images collected from forest surveillance cameras, wildfire monitoring systems, drones, and publicly available image repositories.

Images include:

- Forest scenes containing visible smoke
- Forest scenes without smoke
- Different weather conditions
- Various lighting conditions
- Different smoke densities and distances
- Daytime and nighttime scenes (if available)

Dataset Characteristics

Number of Images

- Approximately 3,000–10,000 images (depending on the selected dataset)

Number of Classes

- 2 Classes

Class 1: Smoke Present
Class 2: No Smoke Present

Image Type

- RGB Images

Image Format

- JPG / PNG

Image Resolution

- Various resolutions

Class Distribution

- To be analysed after dataset selection
- Class balancing techniques may be applied if necessary

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5. Research Questions

The following research questions will guide this study:

RQ1

Can a custom Convolutional Neural Network accurately detect smoke in forest images?

RQ2

How does transfer learning improve smoke detection performance compared to a custom CNN model?

RQ3

Which deep learning model provides the best balance between classification accuracy, precision, recall, and computational efficiency?

RQ4

Can data augmentation techniques improve model generalization and reduce overfitting?

RQ5

Which transfer learning architecture performs best for forest fire smoke detection: MobileNetV2 or ResNet50?

RQ6

How effectively can the trained model distinguish smoke from visually similar environmental conditions such as clouds, fog, or haze?

RQ7

Can Grad-CAM visualizations provide meaningful explanations for the model's predictions?

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6. Expected Results

The expected outcomes of this project include:

Expected Performance

- Classification accuracy greater than 90%
- High precision and recall scores
- Low false positive rate
- Strong generalization on unseen test images

Expected Findings

- Successful development of an automated smoke detection system
- Improved detection performance using transfer learning models
- Better feature extraction capabilities from pretrained networks
- Enhanced model robustness through data augmentation
- Identification of the most suitable architecture for smoke detection

Expected Practical Impact

- Early wildfire detection support
- Reduced monitoring costs
- Faster emergency response
- Potential deployment in surveillance cameras and drone-based monitoring systems

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7. Proposed Methodology

The project will follow a supervised deep learning workflow consisting of data preparation, model development, training, evaluation, and comparison.

Step 1: Data Collection

- Obtain the dataset from Kaggle or Roboflow.
- Verify image quality and class labels.
- Remove duplicate or corrupted images.

Step 2: Data Preprocessing

The following preprocessing techniques will be applied:

- Resize images to 224 × 224 pixels
- Convert images into numerical tensors
- Normalize pixel values to the range [0,1]
- Handle missing or corrupted files
- Encode class labels

Step 3: Data Augmentation

To improve model generalization and reduce overfitting, the following augmentation techniques will be applied:

- Random rotation
- Horizontal flipping
- Zoom augmentation
- Width shifting
- Height shifting
- Brightness adjustment
- Shearing transformations

Step 4: Dataset Splitting

The dataset will be divided into:

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

Step 5: CNN Model Development

A custom CNN architecture will be developed and trained from scratch.

Step 6: Transfer Learning Implementation

Pretrained models will be used as feature extractors and fine-tuned for smoke detection.

Models to be evaluated:

- MobileNetV2

- ResNet50

Step 7: Model Training

Training configuration:

- Optimizer: Adam
- Learning Rate: 0.001
- Batch Size: 32
- Epochs: 20–30
- Loss Function: Binary Crossentropy

Step 8: Model Evaluation

The performance of all models will be compared using multiple evaluation metrics.

Step 9: Explainability Analysis

Grad-CAM visualizations will be generated to identify image regions influencing model predictions.

Step 10: Comparative Analysis

A detailed comparison will be conducted between:

- Custom CNN
- MobileNetV2
- ResNet50

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8. CNN Model Design

Custom CNN Architecture

Input Layer

- Input Shape: 224 × 224 × 3

Convolution Block 1

- Conv2D (32 Filters, 3×3 Kernel)
- ReLU Activation
- MaxPooling2D (2×2)

Convolution Block 2

- Conv2D (64 Filters, 3×3 Kernel)
- ReLU Activation
- MaxPooling2D (2×2)

Convolution Block 3

- Conv2D (128 Filters, 3×3 Kernel)
- ReLU Activation
- MaxPooling2D (2×2)

Flatten Layer

- Converts feature maps into a one-dimensional vector

Fully Connected Layer

- Dense Layer (128 Neurons)
- ReLU Activation
- Dropout (0.5)

Output Layer

- Dense Layer (1 Neuron)
- Sigmoid Activation

Model Summary

Input → Conv32 → Pool → Conv64 → Pool → Conv128 → Pool → Flatten → Dense128 → Dropout → Output

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9. Transfer Learning Models

To improve performance and leverage pretrained knowledge, transfer learning models trained on ImageNet will be used.

Model 1: MobileNetV2

Advantages

- Lightweight architecture
- Fast training and inference
- Suitable for real-time deployment

Implementation

- Load pretrained ImageNet weights
- Freeze initial layers
- Fine-tune final layers
- Replace classification head with binary classifier

Model 2: ResNet50

Advantages

- Deep architecture with residual connections
- Strong feature extraction capability
- High classification accuracy

Implementation

- Load pretrained ImageNet weights
- Replace top classification layers
- Fine-tune deeper layers
- Train on smoke detection dataset

Model Comparison Objectives

Compare:

- Accuracy
- Precision
- Recall
- F1-Score
- Training Time
- Model Size
- Computational Efficiency

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10. Evaluation Metrics

The following evaluation metrics will be used to assess model performance:

Accuracy

Measures overall classification correctness.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Measures how many predicted smoke images are actually smoke.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Measures how many actual smoke images are correctly detected.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

Harmonic mean of precision and recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

ROC-AUC Score

Measures classification capability across different thresholds.

Confusion Matrix

Provides detailed classification results:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

Training and Validation Loss

Used to monitor convergence and overfitting.

11. Expected Figures

The following figures are expected to be generated during the project:

Figure 1

Sample Images from Dataset

Figure 2

Class Distribution Visualization

Figure 3

Data Augmentation Examples

Figure 4

Custom CNN Architecture Diagram

Figure 5

MobileNetV2 Architecture Overview

Figure 6

ResNet50 Architecture Overview

Figure 7

Training Accuracy Curve

Figure 8

Validation Accuracy Curve

Figure 9

Training Loss Curve

Figure 10

Validation Loss Curve

Figure 11

Confusion Matrix for Custom CNN

Figure 12

Confusion Matrix for MobileNetV2

Figure 13

Confusion Matrix for ResNet50

Figure 14

ROC Curve Comparison

Figure 15

Grad-CAM Visualization Examples

Figure 16

Model Performance Comparison Chart

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12. Expected Tables

Table 1

Dataset Summary

Attribute	Description
Dataset Name	Forest Fire Smoke Dataset
Number of Images	To be determined
Number of Classes	2
Image Type	RGB
Image Format	JPG/PNG

Table 2

Class Distribution

Class	Number of Images
Smoke	TBD
No Smoke	TBD

Table 3

Data Split Summary

Dataset Split	Percentage
Training	70%

Dataset Split	Percentage
Validation	15%
Testing	15%

Table 4

Hyperparameter Settings

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	20–30

Table 5

Custom CNN Performance Results

Table 6

MobileNetV2 Performance Results

Table 7

ResNet50 Performance Results

Table 8

Model Comparison Results

Model	Accuracy	Precision	Recall	F1-Score
CNN	TBD	TBD	TBD	TBD
MobileNetV2	TBD	TBD	TBD	TBD
ResNet50	TBD	TBD	TBD	TBD

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13. Expected Contributions

- Development of an automated forest fire smoke detection system using deep learning.
- Comparison between a custom CNN and transfer learning approaches.

- Evaluation of MobileNetV2 and ResNet50 for wildfire monitoring applications.
- Investigation of data augmentation techniques for improving model performance.
- Use of Grad-CAM visualizations to improve model interpretability.
- Potential application in real-time forest surveillance and early wildfire warning systems.